

[8] A Course-Specific Agentic RAG Chatbot for IT Student Support: Architecture, Local Deployment, and Preliminary Evaluation

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1-Minute Executive Takeaway

Key Finding: Confirms on-premise university RAG satisfies privacy and compliance, but underscores the need for hardware-efficient inference.

Core Research Objective

To design, deploy, and evaluate an on-premise agentic RAG chatbot for course-specific IT student tutoring at Hai Phong University.

Methodology & Architecture

Deployed an open-weight language model on a local institutional server, indexing syllabus and lecture documents to support student Q&A interactions.

Key Empirical Findings & Evidence

Proved that local institutional deployments effectively resolve student data privacy concerns and institutional compliance mandates, but identified high server response latency during peak student usage hours.

Direct Relevance to Our Kwiz Paper

Cited in Section 2.3. While Tran et al. focused on conversational student Q&A, our study focuses on instructor-facing structured assessment generation, deterministic schema compilation, and concurrent batch generation.